



INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO



Introduction to Cryptography

Project:TBC16-CTR

May 18, 2026

Please do the following exercises in pairs.

1. Adjustments to the Tiny Block Cipher

Modify your implementation of the TBC as follows.

1. Generate an Sbox and store it in a file with name `XYZ_sbox.txt`, where `XYZ` are the initials of your name.
2. You must have a key generation function to randomly generate a key. The key must be encoded in base64 and store in a textfile.
3. The tiny block cipher (TBC) encryption must be implemented in a function that receives as parameters: the key K , and a block of plaintext M . The output must be a block of ciphertext C .
4. Inside the function TBC encryption, you must include a fixed permutation P and read the Sbox from the textfile, without the intervention of the user. Also you must include the key schedule function inside the this function.

2. Adding the CTR mode of operation

Modify your implementation of the CTR mode of operation as follows.

1. Your implementation of CTR mode encryption must be in a function that receives as parameters the filename containing the plaintext and a the filename containing the random key K . The output must be the ciphertext and the counter used to encrypt the plaintext.
2. The filename of the plaintext and the key must be given by the user.
3. Inside the CTR mode encryption, you must generate at random the counter. Remember that the counter is the concatenation of C_0, C_1 (see lab 08). Also here you must call the procedure TBC encryption to process every block of plaintext.

4. The ciphertext and the counter must be encoded in base 64 and store in the same textfile. The filename must be chosen by the user.
5. Your implementation of CTR mode decryption must be in a function that receives as parameters the filename containing the ciphertext and the counter. The output must be the recovered plaintext and the counter used to encrypt the plaintext.

3. Testing your implementation

To test and show your implementation you must proceed as follows

1. One student in the team will play the role of Alice and the other one will play the role of Bob.
2. Both students must configure the TBC16-CTR they are going to use, with the same Sbox and permutation.
3. Alice will generate a random key K and encrypt a **plaintext of at least 1Kb** using TBC16-CTR to obtain the ciphertext file. Then she will share the K and the ciphertext file with Bob.
4. Bob will use the given key K and the ciphertext file to recover the plaintext.